



**UNIVERSITY
OF LONDON**

Module Name: Computer Security (Midterm Coursework)

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PART A: RESEARCH ON 5 RECENT MALWARE

MALWARE 1: RHYSIDA RANSOMWARE

Malware Type: Ransomware

Date of First Incident: May 2023

Description: Rhysida ransomware emerged in May 2023 with the launch of their victim support chat portal on the dark web. The group, masquerading as a “cybersecurity team,” claims to highlight potential security vulnerabilities in targeted systems. While the specific targets of Rhysida remain unclear, the malware is primarily deployed through Cobalt Strike frameworks and phishing campaigns.

Rhysida employs modern multi-extortion tactics, threatening victims with the public release of exfiltrated data. Upon deployment, it displays a cmd.exe window, traverses all local files, and instructs victims to contact the attackers via a TOR-based portal. Payments are demanded in Bitcoin (BTC), with instructions provided for the transaction. The ransomware uses the ChaCha20 algorithm for file encryption and places ransom notes as PDF documents in affected folders.

Protection Measures: To protect against Rhysida, implement robust anti-malware software, monitor network traffic for unusual patterns, conduct regular security audits, educate employees on cybersecurity best practices, and maintain a comprehensive backup and recovery plan.

Coping with Infection:

1. **Isolation:** Immediately isolate the infected system by disconnecting it from the network to prevent further spread.
2. **Restoration:** Restore systems and files from recent, clean backups stored in a secure, offsite location. Ensure backup integrity through regular verification.
3. **Malware Removal:** Use reputable anti-malware tools to detect and remove Rhysida from infected systems. Follow the software’s instructions for complete eradication and perform subsequent scans to ensure thorough removal.

Source: <https://www.trendmicro.com/vinfo/sg/security/news/ransomware-spotlight/ransomware-spotlight-rhysida>

MALWARE 2: HERMETICWIPER

Malware Type: Destructive or Wiper Malware

Date of First Incident: 23 Feb 2024

Description: HermeticWiper is a targeted wiper malware observed in Ukraine, Latvia, and Lithuania, designed to disrupt government infrastructure. It compromises systems via external-facing servers and uses compromised identities for lateral movement. It requires privileged admin rights to render

systems unbootable by altering boot records, deleting configurations, and erasing backups. Unlike mass-scale attacks, it focuses on targeted entities without widespread distribution mechanisms.

Protection Measures:

1. **Endpoint Privileged Access Controls:** Remove local admin rights and implement strict credential management to prevent unauthorized access and lateral movement.
2. **Identity and Credential Protection:** Safeguard high-privilege credentials, such as Active Directory accounts, to mitigate the risk of exploitation and unauthorized access.
3. **Network Monitoring and Auditing:** Regularly monitor network traffic and conduct audits to detect unusual activities and potential signs of compromise.

Coping Methods if Infected:

1. **Isolate Infected Systems:** Immediately disconnect compromised systems from the network to prevent further spread.
2. **Restore from Backups:** Use recent, clean backups stored securely to restore affected systems and data.
3. **Use Anti-Malware Tools:** Employ reputable anti-malware software to scan and remove HermeticWiper from infected systems thoroughly.

Source: <https://www.cyberark.com/resources/blog/hermeticwiper-what-we-know-about-new-malware-targeting-ukrainian-infrastructure-thus-far>

MALWARE 3 – QUITERAT

Malware Type: Remote Access Trojans, RATs

Date of First Incident: 25 August 2023

Description: A new strain of malware attributed to the Lazarus hacking group, allegedly backed by the North Korean government, targeted healthcare entities and internet backbone infrastructure in Europe and the United States. This malware exploits vulnerabilities in ManageEngine ServiceDesk (CVE-2022-47966) to deploy QuiteRAT, a smaller, more elusive variant of previous Lazarus malware like MagicRAT. QuiteRAT allows for data gathering and can "sleep" to remain undetected on compromised networks. Additionally, Lazarus introduced CollectionRAT, which shares similar capabilities for remote access and command execution.

Protection Measures:

1. **Patch Vulnerabilities:** Ensure all software, including ManageEngine ServiceDesk, is promptly patched to mitigate known vulnerabilities like CVE-2022-47966.
2. **Use Endpoint Security:** Employ advanced endpoint security solutions that can detect and block known malware variants, including RATs like QuiteRAT and CollectionRAT.
3. **Network Monitoring:** Implement robust network monitoring to detect unusual or suspicious network activities indicative of malware infections.

4. **Access Control:** Enforce strict access control policies to limit privileged access, reducing the potential impact of compromised credentials.

Coping Methods if Infected:

1. **Isolate Infected Systems:** Immediately isolate compromised systems from the network to prevent further spread of the malware.
2. **Restore from Backups:** If available, restore affected systems and data from clean backups stored in secure, off-network locations.
3. **Deploy Anti-Malware Tools:** Use reputable anti-malware software to scan and remove the malware from infected systems. Follow best practices for malware removal and system recovery.
4. **Incident Response:** Engage incident response teams to investigate the extent of the infection, identify compromised data, and implement containment and recovery strategies.

Source: <https://thehackernews.com/2023/09/researchers-warn-of-cyber-weapons-used.html>

MALWARE 4 – DURIAN BEACON

Malware Type: Backdoor malware

Date of First Incident: August and November 2023

Description: Kimsuky, a North Korean threat group also known as APT43, has deployed a new sophisticated malware called Durian in targeted cyberattacks against South Korean cryptocurrency firms. Durian operates as a backdoor with extensive capabilities, including command execution, file downloads, and data exfiltration. The malware uses legitimate South Korean software for initial infiltration, followed by the installation of additional malware for persistence. Durian itself, based on the Golang programming language, facilitates the deployment of malicious tools such as AppleSeed and LazyLoad, a custom proxy tool designed to steal browser-stored data like cookies and login credentials.

Protection Measures:

1. **Patch Management:** Keep software and systems updated to mitigate vulnerabilities used for initial infiltration.
2. **Endpoint Security:** Implement robust endpoint security solutions to detect and block backdoor malware like Durian.
3. **Network Monitoring:** Monitor network traffic for unusual patterns or connections indicative of malware activity.

Coping Methods if Infected:

1. **Isolate and Investigate:** Immediately isolate infected systems to prevent further spread and engage in thorough forensic investigation.

2. **Remove Malware:** Use reputable anti-malware tools to scan and remove Durian and associated malware from affected systems.
3. **Enhance Security Posture:** Review and enhance cybersecurity measures to prevent future attacks, including user awareness training and access control policies.

Source: <https://www.scmagazine.com/brief/kimsuky-observed-deploying-new-durian-malware>

MALWARE 5 – SPINOK ANDROID MALWARE

Malware Type: Android Spyware

Date of First Incident: May 2023

Description: SpinOk malware infiltrated popular Android apps, including those with millions of downloads from Google Play. It masquerades as a benign advertisement SDK offering minigames and rewards but secretly steals sensitive data such as passwords, credit card information, and cryptocurrency details. It avoids detection by checking sensor data and operates stealthily in the background.

Protection Measures:

1. **Update or Uninstall Affected Apps:** Ensure all apps are updated to clean versions available on Google Play. Uninstall apps if clean versions are not available.
2. **Use Mobile Antivirus Tools:** Scan devices with reputable mobile antivirus software to detect and remove any remaining spyware.
3. **Enable Google Play Protect:** Utilize Google Play Protect, which warns users about apps exhibiting malicious behavior, even if downloaded from sources other than Google Play.

Coping Methods if Infected:

1. **Remove the App:** Immediately uninstall the affected app from your device.
2. **Scan for Residual Spyware:** Conduct a thorough scan using mobile antivirus tools to eliminate any remaining spyware.
3. **Monitor Financial Accounts:** Monitor your accounts for any unauthorized transactions and take appropriate action, such as changing passwords and notifying financial institutions.

Source: <https://www.bleepingcomputer.com/news/security/android-apps-with-spyware-installed-421-million-times-from-google-play/>